

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 17/01/2026**
Time: 3 Hrs **GTM - 17** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. In a Young's double slit experiment, 12 fringes are observed in a certain segment of the screen when light of wave length 600 nm is used. If the wave length of light is changed to 400 nm, the number of fringes observed in the same segment of the screen is given by

   A) 12 B) 18 C) 24 D) 30

2. Radio waves received by a radio telescope from distant star may have a wavelength of about 0.20 m. If the speed of the wave is $3\times10^{8}\,ms^{-1}$, then frequency of the wave will be

   A) $1.5\times10^{9}\,Hz$ B) $1.5\times10^{8}\,Hz$ C) $1.5\times10^{3}\,Hz$ D) 135 Hz

Rankers Academy JEE
EDUCATIONAL INSTITUTIONS
APEX
JEE-MAIN 2026
For More Join: @IITJEE_Advanc

3. Statement-1: In a conductor, free electrons keep on moving but no magnetic force acts on a conductor placed in the magnetic field produced by electrons.

Statement-2: Force on free electron due to external magnetic field always acts perpendicular to its direction of motion.

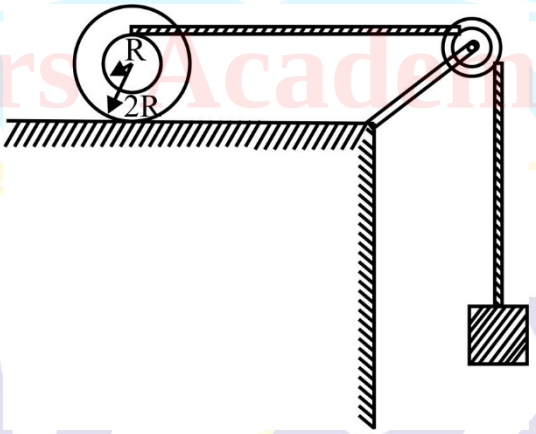
A) Statement-1 is True, statement-2 is True; statement-2 is a correct explanation for statement-1

B) Statement-1 is True, statement-2 is True; statement-2 is not a correct explanation for statement-1

C) Statement-1 is True, statement-2 is False

D) Statement-1 is False, statement-2 is True

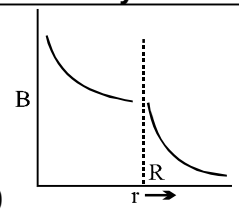
4. In the figure shown mass of both, the spherical body and block is m. Moment of inertia of the spherical body about center of mass is $2mR^2$. The spherical body rolls on the horizontal surface. The ratio of kinetic energy of the spherical body to that of block at any instant is (friction is sufficient to have pure rolling for spherical body. Pulley and string are massless and friction less.)

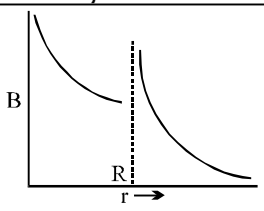


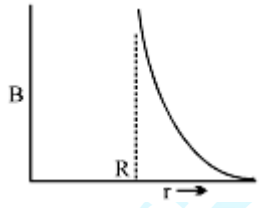
A) $\frac{3}{4}$ B) $\frac{1}{3}$ C) $\frac{2}{3}$ D) $\frac{1}{2}$

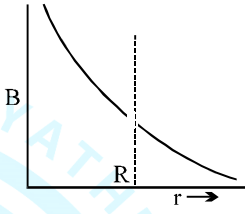
5. A thin, straight conductor lies along the axis of a hollow conductor of radius R. The two carry equal currents in the same direction. The magnetic field B is plotted against the distance r from the axis. Which of the following best represents the resulting curve?

For More Join: @IITJEE_Advanc

A) 

B) 

C) 

D) 

6. Statement I :- The total kinetic energy of all the molecules of a given mass of an ideal gas is 1.5 times the product of its pressure and volume.

Statement II:- The molecules of a gas collide with each other and the velocities of the molecules change due to the collision .

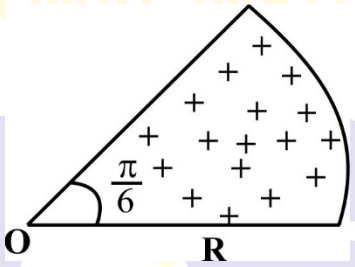
A) Statement-1 is False, statement-2 is True

B) Statement-1 is False, statement-2 is False

C) Statement-1 is True, statement-2 is False

D) Statement-1 is True, statement-2 is True

7. The diagram shows a part of disc of radius R carrying uniformly distributed charge of surface charge density σ. Electric potential at the centre O of disc is



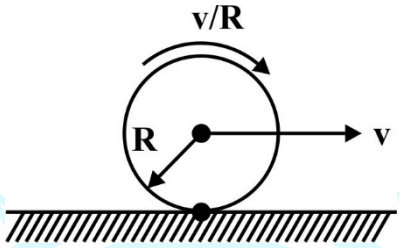
A) $\frac{\sigma R}{2\varepsilon_0}$ B) $\frac{\sigma R}{3\varepsilon_0}$ C) $\frac{\sigma R}{24\varepsilon_0}$ D) $\frac{\sigma R}{6\varepsilon_0}$

8. A particle moving in the positive x-direction has initial velocity v_0. The particle undergoes retardation kv^2, where v is its instantaneous velocity. The velocity of the particle as a function of time (t) is given by

A) $v=\frac{v_0}{(1+kv_0t)}$ B) $v=\frac{2v_0}{1+kt}$ C) $v=\frac{v_0}{kt}$ D) $v=\frac{v_0}{(1+k^2v_0^2t)}$

For More Join: @IITJEE_Advanc

9. A disc is performing pure rolling on a smooth stationary surface with constant angular velocity as shown in figure. At any instant, for the lower most point of the disc



A) velocity is v, acceleration is zero

B) velocity is zero, acceleration is zero

C) velocity is v, acceleration is $\frac{v^2}{R}$

D) velocity is zero, acceleration is $\frac{v^2}{R}$

10. Two point dipoles $p\hat{k}$ and $\frac{p}{2}\hat{k}$ are located at $(0,0,0)$ and $(1m,0,2m)$ respectively. The resultant electric field due to the two dipole at the point $(1m,0,0)$ is___

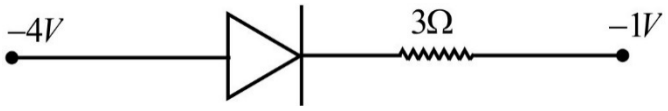
A) $\frac{9p}{32\pi\varepsilon_0}\hat{k}$ B) $-\frac{7p}{32\pi\varepsilon_0}\hat{k}$ C) $\frac{7p}{32\pi\varepsilon_0}\hat{k}$ D) $-\frac{11p}{32\pi\varepsilon_0}\hat{k}$

11. A steady current is flowing in a circular coil of radius R, made up of a thin conducting wire. The magnetic field at the centre of the loop is B_L. Now, a circular loop of radius $\frac{R}{n}$ is made from the same wire without changing its length, by unfolding and refolding the loop, and the same current is passed through it. If new magnetic field at the centre of the coil is B_C, then the ratio $\frac{B_L}{B_C}$ is (n is number of turns)

A) $1:n^2$ B) $n^{3/2}:1$ C) $n:1$ D) $n^2:1$

For More Join: @IITJEE_Advanc

12. The value of current in the following diagram is (diode assumed to be ideal one)



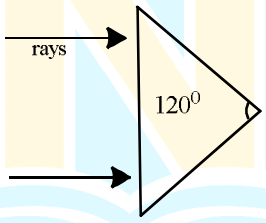
A) 0.1 amp B) 0.01 amp C) 1 amp D) zero

13. The temperature of the surface of the sun is measured with

A) Platinum thermometer B) Gas thermometer

C) Pyrometer D) Vapour pressure thermometer

14. An isosceles prism of angle 120^0 has refractive index 1.44. Two parallel rays of monochromatic light enter the prism parallel to each other in air as shown in figure. The rays emerging from the opposite face



A) are parallel to each other

B) are diverging

C) make an angle $2\left\{\sin^{-1}(0.72)-30^0\right\}$ with each other

D) make an angle $2\sin^{-1}(0.72)$ with each other

15. The dimensional formula of $a\times b$ in the relation $P=\dfrac{b-x^2}{at}$ is, where P is power. x is distance and t is time.

A) $\left[M^{-1}L^2T^2\right]$ B) $\left[M\,L^2T^2\right]$ C) $\left[ML^2T^{-2}\right]$ D) $\left[M^{-1}LT^2\right]$

16. A body cools down from 50^0C to 45^0C in 5 min and 45^0C to 40^0C is 8 min. The temperature of the surrounding closer to

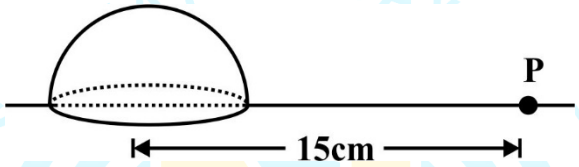
A) 34^0C B) 24^0C C) 39^0C D) 20^0C

17. A electron having kinetic energy T is moving in a circular orbit of radius R perpendicular to a uniform magnetic field induction $\vec{B}$. If kinetic energy is doubled and magnetic field induction tripled, then the radius will become

For More Join: @IITJEE_Advanc

A) $\frac{3R}{2}$ B) $\sqrt{\frac{3}{2}}R$ C) $\sqrt{\frac{2}{9}}R$ D) $\sqrt{\frac{4}{3}}R$

18. Figure shows a solid hemisphere with a charge of 5 nC distributed uniformly throughout its volume. The hemisphere lies on a plane and point P is located on the plane, along a radial line from the centre of curvature at distance 15 cm. The electric potential at point P due to the hemisphere, is:



A) 150 V B) 300V C) 450 V D) 600 V

19. The potential energy of a particle of mass 1 kg in motion along the x-axis is given by $U=(10-10\cos 6x)J$. The period of small oscillations is $(\pi^2 \approx 10)$

A) $\frac{1}{3}s$ B) $\frac{3}{2}s$ C) $\frac{2}{3}s$ D) $\frac{1}{4}s$

20. The ratio of de Broglie wavelengths of molecules of hydrogen and helium which are at temperature 27^0C and 127^0C, respectively is

A) $\frac{1}{2}$ B) $\sqrt{\frac{3}{8}}$ C) $\sqrt{\frac{8}{3}}$ D) 1

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. An electron makes 3×10^5 revolutions per second in a circle of radius 0.5 angstrom. The magnetic field B at the centre of the circle is $n\times10^{-10}T$. Then find the value of n (nearest integer)

For More Join: @IITJEE_Advanc

22. Find the maximum speed at which a car can turn round a curve of 30 m radius on a level road of the coefficient of friction between the tyres and the road is 0.4 [acceleration due to gravity $=10\,m/s^2$]

23. When the energy of the incident radiation is increased by 20%, the kinetic energy of the photoelectrons increase from $0.5eV$ to $0.8eV$. The work function of the metal (in eV) is___

24. The binding energy of an electron in the ground state of the He atom is equal to $E_0 = 24.6eV$. Find the energy required (in eV) to remove both electrons from the He atom.

25. A planet revolves about the sun in elliptical orbit of semimajor axis $2\times10^{12}\,m$. The areal velocity of the planet when it is nearest to the sun is $4.4\times10^{15}\,m^2/s$. The least distance between planet and the sun is$1.8\times10^{12}\,m$. Then find the minimum speed of the planet in km/s

CHEMISTRY MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Consider the following reduction process

$Zn^{2+} + 2e^- \rightarrow Zn(s); E^0 = -0.76V$

$Ca^{2+} + 2e^- \rightarrow Ca(s); E^0 = -2.87V$

$Mg^{2+} + 2e^- \rightarrow Mg(s); E^0 = -2.36V$

$Ni^{2+} + 2e^- \rightarrow Ni(s); E^0 = -0.25V$

The reducing power of the metals increases in the order

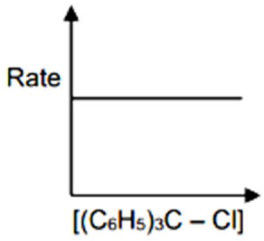
A) $Ca < Zn < Mg < Ni$ B) $Ni < Zn < Mg < Ca$

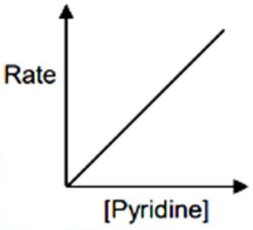
C) $Zn > Mg < Ni < Ca$ D) $Ca < Mg < Zn < Ni$

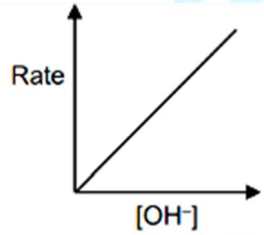
For More Join: @IITJEE_Advanc

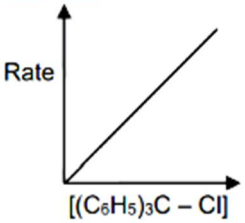
27. The graph which represents the following reaction

$$(C_6H_5)_3C-Cl \xrightarrow[\text{Pyridine}]{OH^-} (C_6H_5)_3C-OH$$

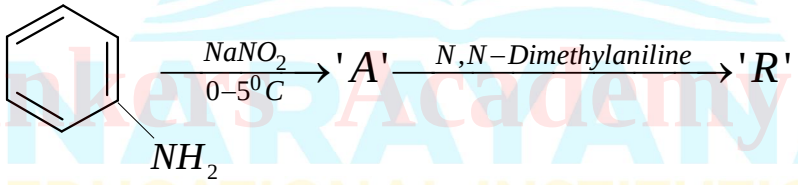
A) 

B) 

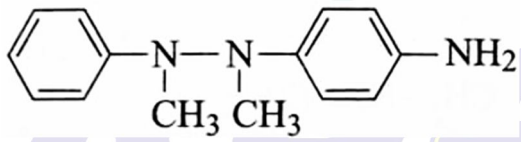
C) 

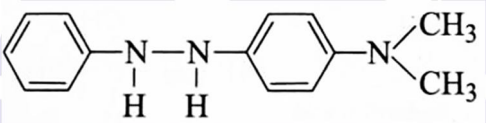
D) 

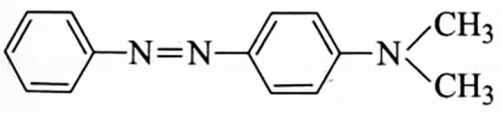
28. Consider the following sequence of reactions:

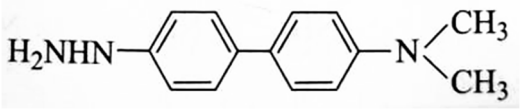


The product R is

A) 

B) 

C) 

D) 

For More Join: @IITJEE_Advanc

29. Given below are two statements:

Statement-I : The orbitals having same energy are called as degenerate orbitals

Statement-II : In hydrogen atom, 3p and 3d orbitals are not degenerate orbitals

In the light of the above statement, choose the most appropriate answer from the options given

A) Statement-I is true but statement-II is false

B) Both Statement-I and Statement-II are true

C) Both Statement-I and Statement-II are false

D) Statement-I is false but Statement-II is true

30. Hybridization in 1, 3–butadiene has

A) sp and sp^2 hybridized Carbon atoms

B) sp, sp^2 and sp^3 hybridized Carbon atoms

C) Only sp^2 hybridized Carbon atoms

D) Only sp hybridized Carbon atoms

31. The exact volumes of $1M\ NaOH$ solution required for complete neutralisation 50mL of $1M\ H_3PO_3$ solution and 100 mL of $2M\ H_3PO_2$ solution, respectively, are

A) 100 mL and 100 mL B) 100 mL and 50 mL

C) 100 mL and 200 mL D) 50 mL and 50 mL

32. Identify the species having effectively one delocalised π-system and the maximum number of canonical (resonance) forms from the following:

A) CO_3^{2-} B) O_3 C) SO_2 D) NO_2

33. Match the List-I with List-II

List-I (Cations)	**List-II (Group reagents)**
$A \rightarrow Pb^{2+}, Cu^{2+}$	i) H_2S gas in presence of dilute HCl
$B \rightarrow Al^{3+}, Fe^{3+}$	ii) $(NH_4)_2CO_3$ in presence of NH_4OH
$C \rightarrow Co^{2+}, Ni^{2+}$	iii) NH_4OH in presence of NH_4Cl
$D \rightarrow Ba^{2+}, Ca^{2+}$	iv) H_2S in presence of NH_4OH

Correct match is

For More Join: @IITJEE_Advanc

A) $A \to iv, B \to ii, C \to iii, D \to i$ B) $A \to i, B \to iii, C \to ii, D \to iv$

C) $A \to iii, B \to i, C \to iv, D \to ii$ D) $A \to i, B \to iii, C \to iv, D \to ii$

34. For reaction; $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_{(g)}$. The value of K_p changes with

A) Catalyst B) Temperature

C) Amounts of H_2 and I_2 D) All the above factors

35. What is the spin-only magnetic moment value (BM) of a divalent metal ion with atomic number 25, in its aqueous solution?

A) 5.92 B) 5.0 C) zero D) 5.26

36. The Empirical formula of compound is CH_2O. If its molecular weight is 180. The molecular formula of the compound is

A) $C_3H_6O_3$ B) $C_4H_8O_4$ C) $C_6H_{12}O_6$ D) $C_5H_{10}O_5$

37. Homoleptic octahedral complexes of a metal ion M^{3+} with three different monodentate ligands L_1, L_2 and L_3 absorb light in the yellow, blue and red regions of the visible spectrum respectively.

Based on this information, the increasing order of ligand field strength is.

A) $L_2 < L_1 < L_3$ B) $L_3 < L_2 < L_1$ C) $L_3 < L_1 < L_2$ D) $L_1 < L_2 < L_3$

38. Cyclohexanecarbaldehyde reacts with which of the following to produce methyl cyclohexane

A) $PhMgBr$ and then $H_3O^{\oplus}$ B) Tollen's reagent

C) Excess ethanol and acid D) Zinc – amalgam and $dil.HCl$

39. Arrange the following in increasing order of reactivity towards nitration

A) *p*-xylene B) bromobenzene C) mesitylene D) nitrobenzene

E) benzene

Choose the correct answer from the options given below

A) $C < D < E < A < B$ B) $D < B < E < A < C$

C) $D < C < E < A < B$ D) $C < D < E < B < A$

For More Join: @IITJEE_Advanc

40. Which one of the following statements is incorrect relation to ionisation enthalpy

A) The 1st ionisation enthalpy of Mg is higher than Al

B) The greatest increase in ionisation enthalpy is experienced on removal of electron from core noble gas configuration

C) End of valence electrons is marked by a big jump in ionisation enthalpy

D) Removal of electron from orbitals bearing lower $'n'$ value is easier than from orbital having higher $'n'$ value

41. Given,

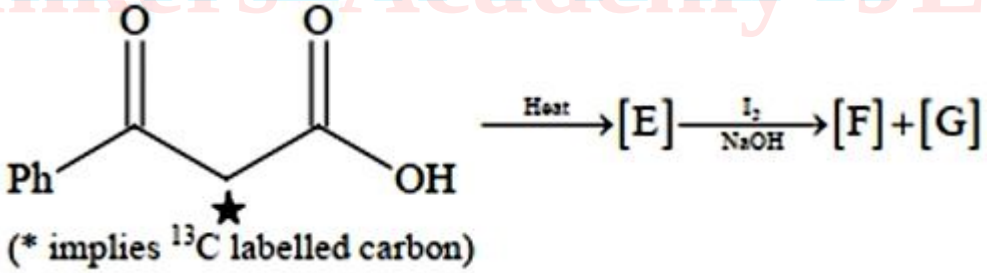
A) $2CO(g)+O_2(g)\rightarrow 2CO_2(g) \quad \Delta H_1^0=-x\,kJ\,mol^{-1}$

B) $C(graphite)+O_2(g)\rightarrow CO_2(g) \quad \Delta H_2^0=-y\,kJ\,mol^{-1}$

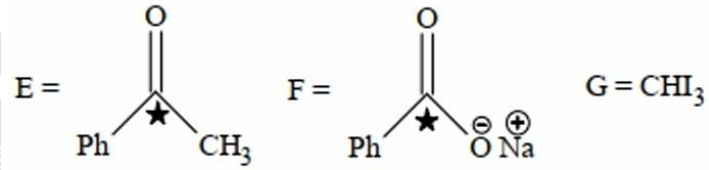
The ΔH^0 for the reaction $C(graphite)+\frac{1}{2}O_2(g)\rightarrow CO(g)$ is___

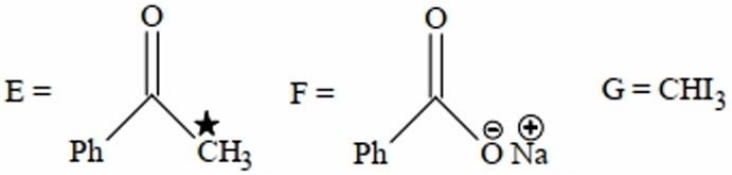
A) $\frac{x+2y}{2}$ B) $\frac{x-2y}{2}$ C) $\frac{2x-y}{2}$ D) $2y-x$

42. In the following reaction sequence, the correct structures of E, F and G are

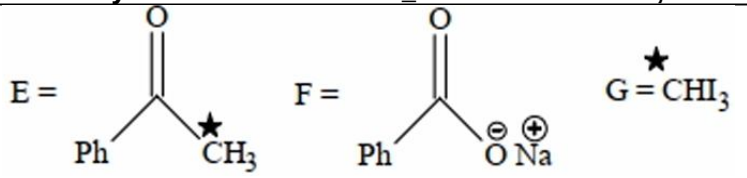


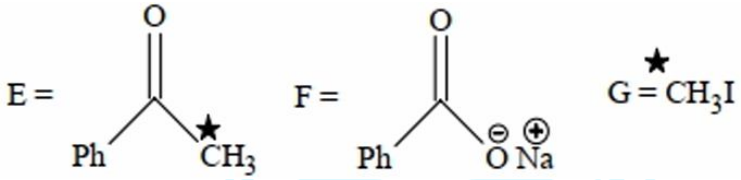
(* implies ^{13}C labelled carbon)

A) 

B) 

For More Join: @IITJEE_Advanc

C) 

D) 

43. Which of the following reaction is not correct

A) $H_3C-CH_2-CH_2-COOC_2H_5 \overset{NaOH}{\rightleftharpoons} CH_3-CH_2-CH_2-COONa+C_2H_5OH$

B) $CH_3-CH_2-CH_2-\underset{\displaystyle OH}{\underset{|}{CH_2}} \xrightarrow[\text{Jones reagent}]{CrO_3,H_2SO_4} H_3C-CH_2-CH_2-COOH$

C) $C_6H_5-CH_2OH \xrightarrow{HBr} C_6H_5-CH_2-Br \xrightarrow{KCN} C_6H_5-CH_2-CN \xrightarrow[H_3O^{\oplus}]{\Delta} C_6H_5-COOH$

D) $CH_2-CH_2-CH_2-CHO \xrightarrow[solution]{Ammonical\ AgNO_3} CH_3-CH_2-CH_2-COOH$

44. Two nucleotides are joined together by a linkage known as:

A) Phosphodiester linkage B) Glycosidic linkage

C) Disulphide linkage D) Peptide linkage

45. Identify the elements X and Y using the ionization energy values given below.

	Ionization energy	**(KJ/mol)**
	1st	2nd
X	495	4563
Y	731	1450

A) $X = Na; Y = Mg$ B) $X = Mg; Y = F$

C) $X = Mg; Y = Na$ D) $X = F; Y = Mg$

For More Join: @IITJEE_Advanc

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. 0.004 M aqueous solution of K_2SO_4 is isotonic with a 0.01 M aqueous solution of glucose at the same temperature. (Assuming that K_2SO_4 dissociates only partially in solution) calculate the percentage dissociation of K_2SO_4.

47. $CH_3-CH=\underset{\underset{Br}{|}}{C}-\underset{\underset{Cl}{|}}{C}=CH-CH_3$

Number of geometrical isomers are possible for this compound is……

48. The decomposition of formic acid on gold surface follows first order kinetics. If the rate constant at 300 K is $1.0\times10^{-3}s^{-1}$ and the activation energy $E_a=11.488\,kJmol^{-1}$ the rate constant at 200K is…..$\times10^{-5}s^{-1}$. (Given: $R=8.314\,J\,mol^{-1}K^{-1}$) (nearest integer)

49. Number of hydrogen atoms per molecule of a hydrocarbon A having 85.8% carbon is…… (Given molar mass of $A=84g\,mol^{-1}$)

50. In Carius method of estimation of halogen, 0.15g of an organic compound gave 0.12 g of $AgBr$. The percentage of bromine in the organic compound is……..%
(Given molar mass of $AgBr=188g/mol$, $Ag=108\,g/mol, Br=80g/mol$)

For More Join: @IITJEE_Advanc

MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Group A contains 7 Boys and 4 Girls, while Group B contains 5 Boys and 4 Girls. The number of ways 4 Boys and 4 Girls can be invited for a picnic if 5 of them must be from Group A and remaining 3 from Group B is

A) 10290 B) 10220 C) 9730 D) 9660

52. If the system of equations $x + 2y - 3z = 2$; $2x + \lambda y + 5z = 5$; $14x + 3y + \mu z = 33$ has infinitely many solutions then the number of the positive integral divisors of $\lambda + \mu$ is

A) 4 B) 5 C) 6 D) 7

53. Let S = $\left\{x \in R : x \geq 0, 2\left|\sqrt{x}-3\right| + \sqrt{x}\left(\sqrt{x}-6\right)+6=0\right\}$ then number of elements in set S is

A) 0 B) 1 C) 2 D) 4

54. Area bounded by the curves $y = e^x, y = \left|e^x - 1\right|$ and x – axis is

A) $ln\,2$ B) $2 - ln\,2$ C) $3 - 2ln\,2$ D) $2 - 2ln\,2$

55. Let e be the eccentricity of hyperbola and $f(e)$ be the eccentricity of its conjugate hyperbola then $\int\left(f(e) + f\left(f(e)\right)\right)de = g(e)$ and $g(1) = \frac{1}{2}$, then $g(e) =$ ____

A) $\frac{1}{2}\sqrt{e^2-1} + \frac{e^2}{2}$ B) $\frac{1}{2}\sqrt{e^2+1} + \frac{e^2}{2}$

C) $\sqrt{e^2-1} + \frac{e^2}{2}$ D) $\sqrt{e^2+1} + \frac{e^2}{2}$

For More Join: @IITJEE_Advanc

56. The point P $(\alpha, \alpha+1)$ will lie on or inside the triangle with sides $3x-2y+6=0$; $x-8y+2=0$; $x+3y-9=0$ then the product of minimum and maximum values of α is_____

A) $-\frac{7}{9}$ B) $-\frac{9}{7}$ C) $-\frac{14}{3}$ D) $-\frac{3}{14}$

57. Let f : $(0,\infty) \to$ R be a function which is differentiable at all points in its domain satisfy $x^2 f'(x) = 2xf(x)+3$ with $f(2) = \frac{39}{2}$ then 3f(3) is____

A) 134 B) 143 C) 413 D) 93

58. If $14 = 6 + \frac{1}{7}(6+\alpha)+\frac{1}{7^2}(6+2\alpha)+\frac{1}{7^3}(6+3\alpha)+.....\infty$ then the value of α is___

A) 35 B) 36 C) 42 D) 48

59. Let m and n be the number of points for which the function f (x) = [x] + $|x-1|$, $-2 < x < 3$ is not continuous and non – differentiable respectively then m + n is ([.] is GIF)

A) 5 B) 4 C) 10 D) 8

60. Let A = $[a_{ij}]$ be square matrix of order 3 with entries {-1 , 0 , 1}. Let E be the event that A is a symmetric matrix then P(E) is

A) $\frac{1}{3}$ B) $\frac{1}{9}$ C) $\frac{1}{27}$ D) $\frac{1}{81}$

61. The position vectors of three vertices of triangle be $3\bar{p}+\bar{q}-2\bar{r}$; $-4\bar{p}+\bar{q}+2\bar{r}$ and $2\bar{p}-\bar{q}+\bar{r}$. If the Position vector of orthocentre and circumcentre be $\frac{\bar{p}+\bar{q}+\bar{r}}{4}$ and $\alpha\bar{p}+\beta\bar{q}+\gamma\bar{r}$ respectively then $3\alpha+4\beta+\gamma$ is

A) 1 B) 6 C) 4 D) 3

For More Join: @IITJEE_Advanc

62. Let $\bar{a} = 2\hat{i} - \hat{j} + \hat{k}$; $\bar{b} = \bar{a} \times \left(\hat{i} - 2\hat{k}\right)$ and $\bar{c} = \bar{b} \times \hat{k}$ then the projection of $\bar{c} + 2\hat{j}$ on $\bar{a}$ is

A) $\frac{5}{\sqrt{6}}$ B) $\frac{7}{\sqrt{6}}$ C) $\frac{10}{\sqrt{6}}$ D) $\frac{11}{\sqrt{6}}$

63. The number of real solutions of $x^2 + 2x - 3 = \min\left\{|x-3|, |x+2|\right\}$ is

A) 1 B) 2 C) 3 D) 4

64. The function f : $(-\infty, 0) \to (-\infty, -1)$ defined by f $(x) = \frac{e^x + e^{-x}}{e^x - e^{-x}}$ is

A) One – One but not onto B) Neither one – one nor onto

C) Onto but not one – one D) Both one – one and onto

65. In an A.P , If $S_{40} = 970$, $S_{12} = 39$ then $S_{30} - S_{10}$ is

A) 385 B) 485 C) 585 D) 685

66. Suppose A and B are the coefficients of 30th and 12th terms respectively in the binomial expansion of $(1+x)^{2n-1}$. If 2A = 5B then the number of positive integral divisors of n is

A) 7 B) 3 C) 5 D) 4

67. Let (2 , 3) be the largest open interval in which the function $f(x) = 2ln(x-2) - 2x^2 + ax + 3$ is Strictly increasing and (b , c) be the largest open interval, in which the function. $g(x) = (x-1)^3 (x+8-a)^2$ is strictly decreasing. Then value of 10(a + b – 3c) is

A) 56 B) 48 C) 62 D) 46

For More Join: @IITJEE_Advanc

68. For some a , b , let f (x) = $\begin{vmatrix} a-\frac{\sin x}{x} & 1 & b \\ a & 1+\frac{\sin x}{x} & b \\ a & 1 & b-\frac{\sin x}{x} \end{vmatrix}$ $x \neq 0$, if $\lim\limits_{x\to 0} f(x) = \lambda_1 + \lambda_2 a + \lambda_3 b$ then $(\lambda_1+\lambda_2+\lambda_3)^2$ is

A) 0 B) 1 C) 4 D) 9

69. Water is pouring into a conical vessel with vertex upwards and closed base at the rate of 3 cubic meters per minute through a tiny hole at the vertex. The radius of the base is 5m and the height is 10m. when the level is 7m from the base, the rate at which water level increases (in m/min) is_____

A) $\frac{18}{7\pi}$ B) $\frac{9\pi}{7}$ C) $\frac{4}{3\pi}$ D) $\frac{9}{7\pi}$

70. If $\alpha > \beta > \gamma > 0$ then the expression $\cot^{-1}\left(\frac{\alpha^2+1}{\alpha-\beta}-\alpha\right)+\cot^{-1}\left(\frac{\beta^2+1}{\beta-\gamma}-\beta\right)+\cot^{-1}\left(\frac{\gamma^2+1}{\gamma-\alpha}-\gamma\right)$ is equal to

A) $-\pi$ B) 0

C) π D) $\frac{\pi}{2}-(\alpha+\beta+\gamma)$

For More Join: @IITJEE_Advanc

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. Let P be image of point Q (1, 2, –3) with respect to the line $L: \frac{x+1}{2} = \frac{y-3}{-2} = \frac{z}{-1}$ and $R(-1, a, b)$ be a point on L = 0. Then square of area of ΔPQR is____

72. $\int \frac{2x^2+5x+9}{\sqrt{x^2+x+1}}dx = x\sqrt{x^2+x+1} + \alpha\sqrt{x^2+x+1} + \beta \log_e \left| x + \frac{1}{2} + \sqrt{x^2+x+1} \right| + c$, where c is constant of integration then the value of $6\alpha + 4\beta$ is____

73. Let $y = y(x)$ be solution of differential equation $\frac{dy}{dx} + x.\sin 2y = x^3.\cos^2 y, x \in \left[0, \frac{\pi}{2}\right)$. If y (0) = $\frac{\pi}{4}$ then $2.\lim_{x\to 0} \tan y$ is____

74. Find total number of functions f : { 1 , 2 , 3 , 4} → { 1 , 2 , 3 , 4 , 5 , 6 } such that f (1) + f (2) = f (3) is____

75. Let hyperbola H : $\frac{x^2}{a^2} - y^2 = 1$ and ellipse E : $3x^2 + 4y^2 = 12$ such that lengths of latusrectum of H and E are equal. If e_H and e_E are eccentricities of H and E respectively. The value of $12\left(e_H^2 + e_E^2\right)$ is____

For More Join: @IITJEE_Advanc